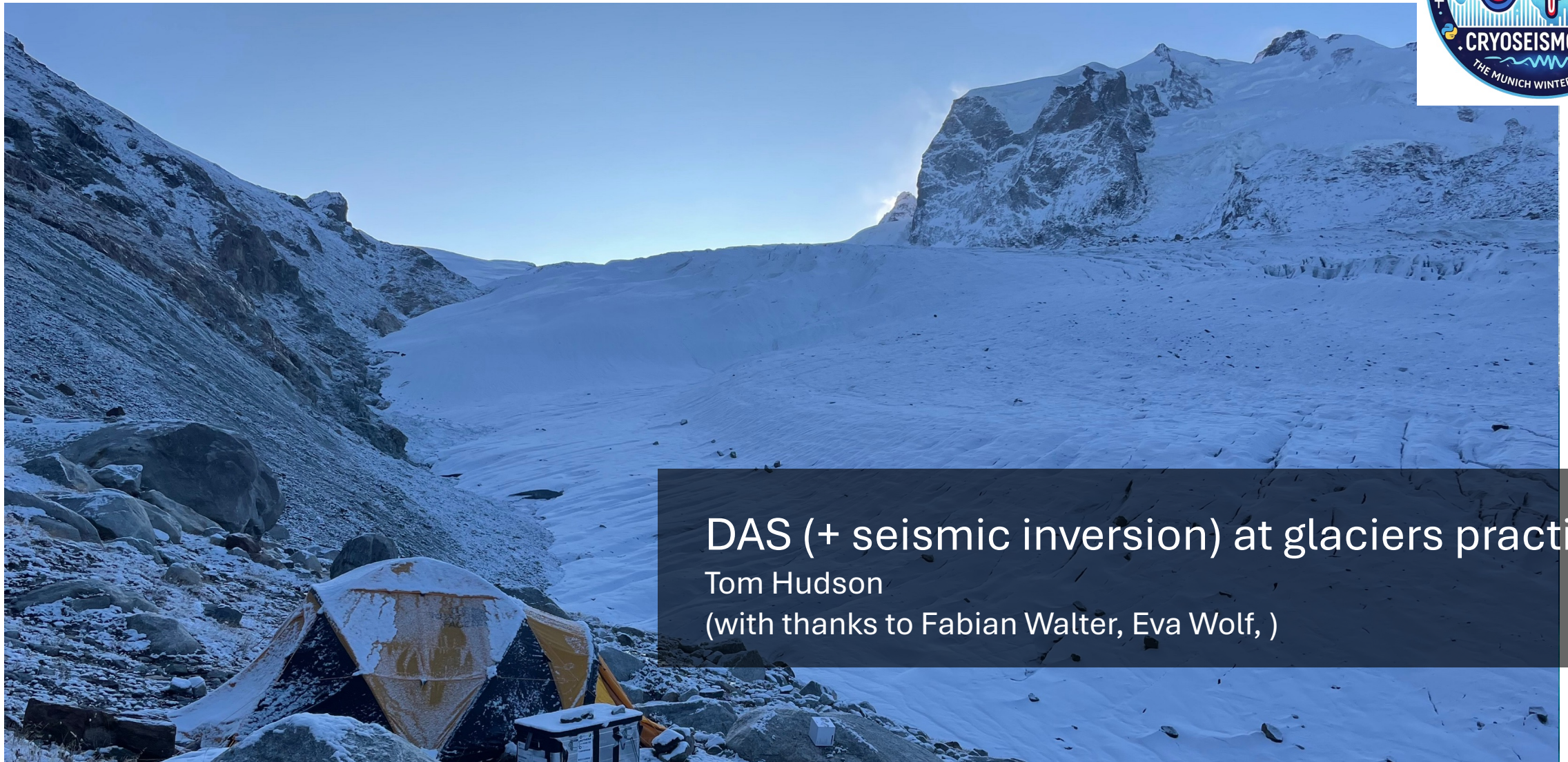




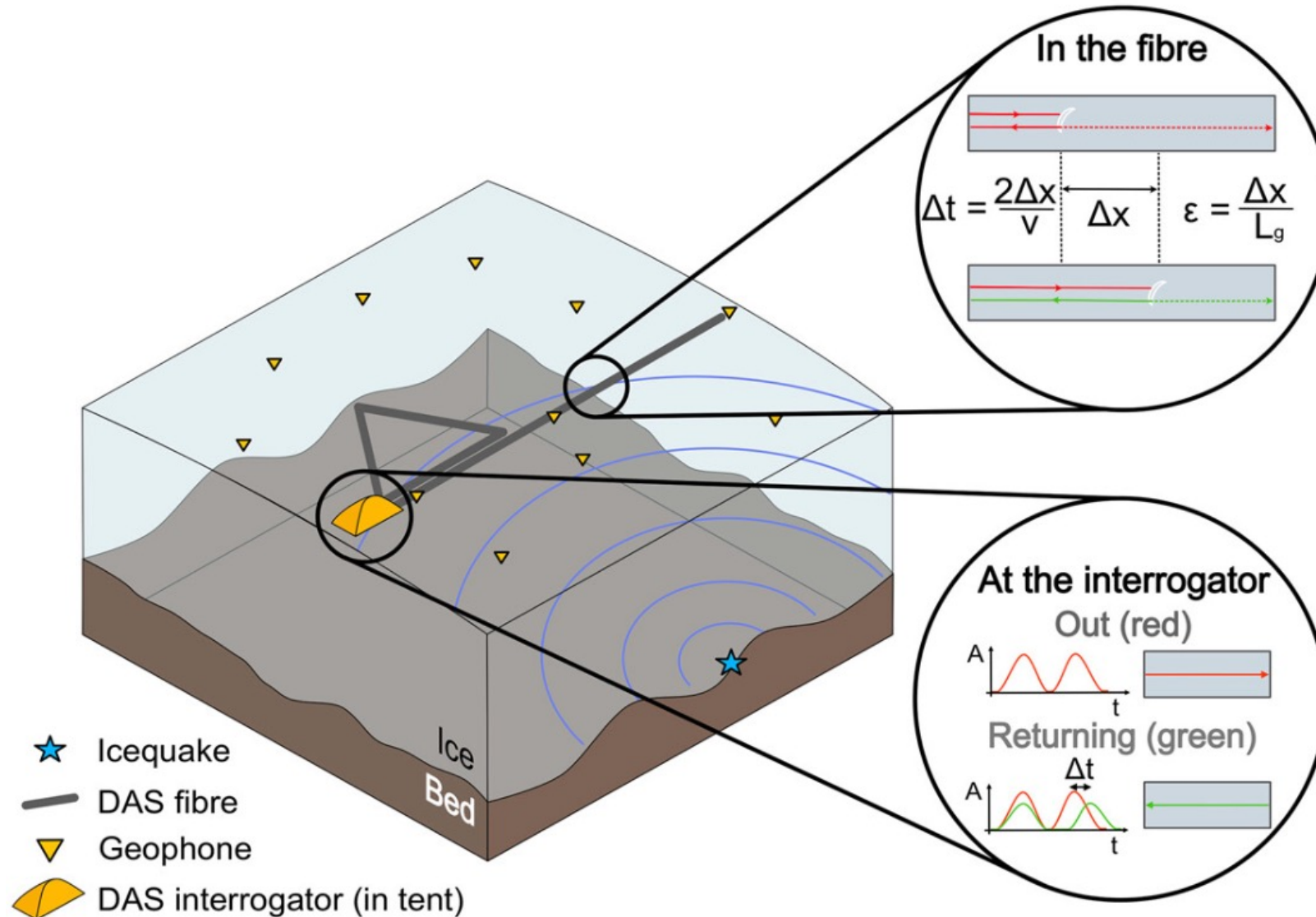
DAS (+ seismic inversion) at glaciers practical

Tom Hudson
(with thanks to Fabian Walter, Eva Wolf,)

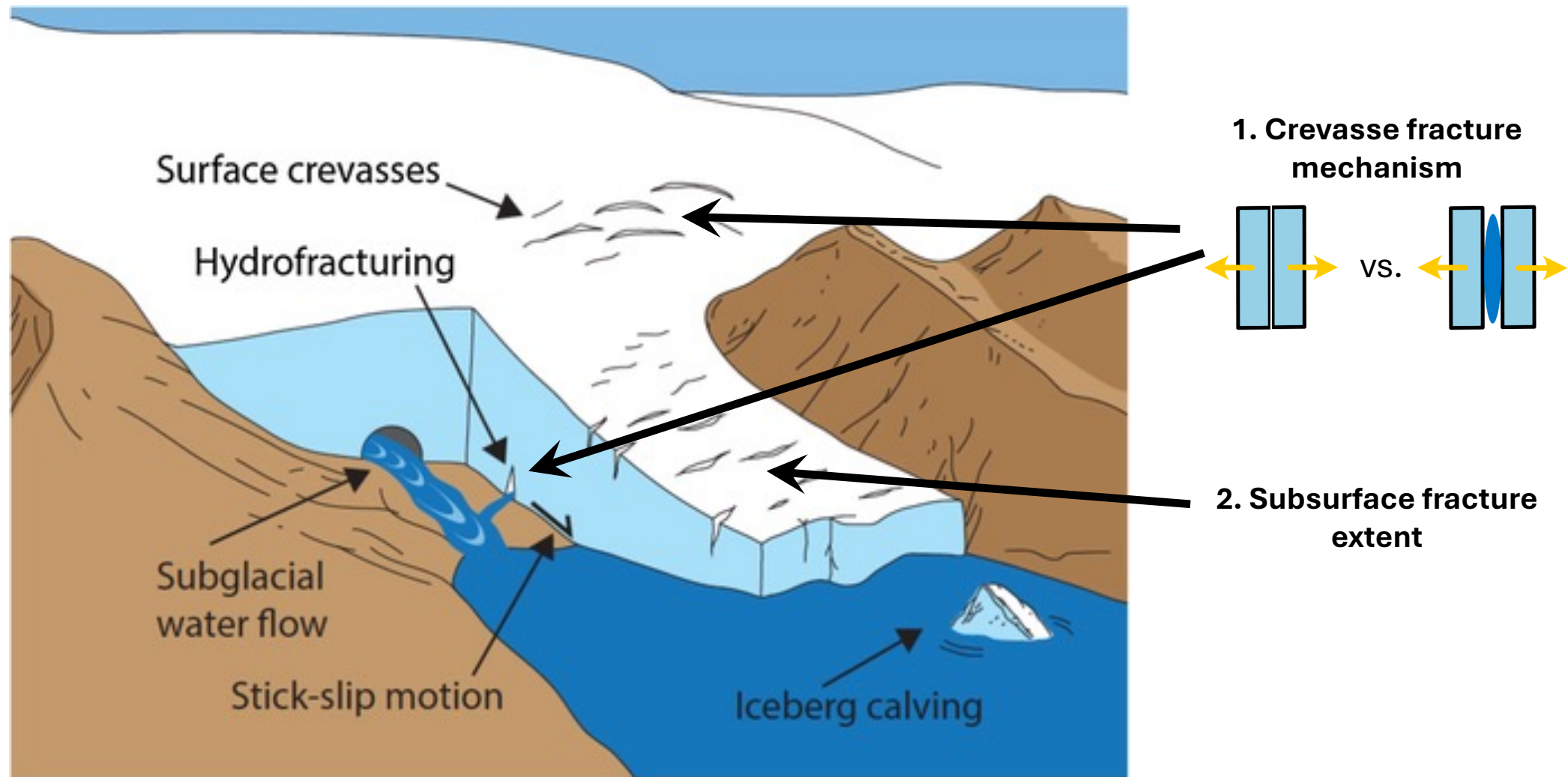
Overview of practical

1. Some context
2. The dataset
3. Choice of two different inversion practicals
 - Source inversion
 - “Tomographic” inversion

Distributed Acoustic Sensing (DAS)



Two glacier processes that we are interested in



Podolskiy and Walter (2016)

The dataset

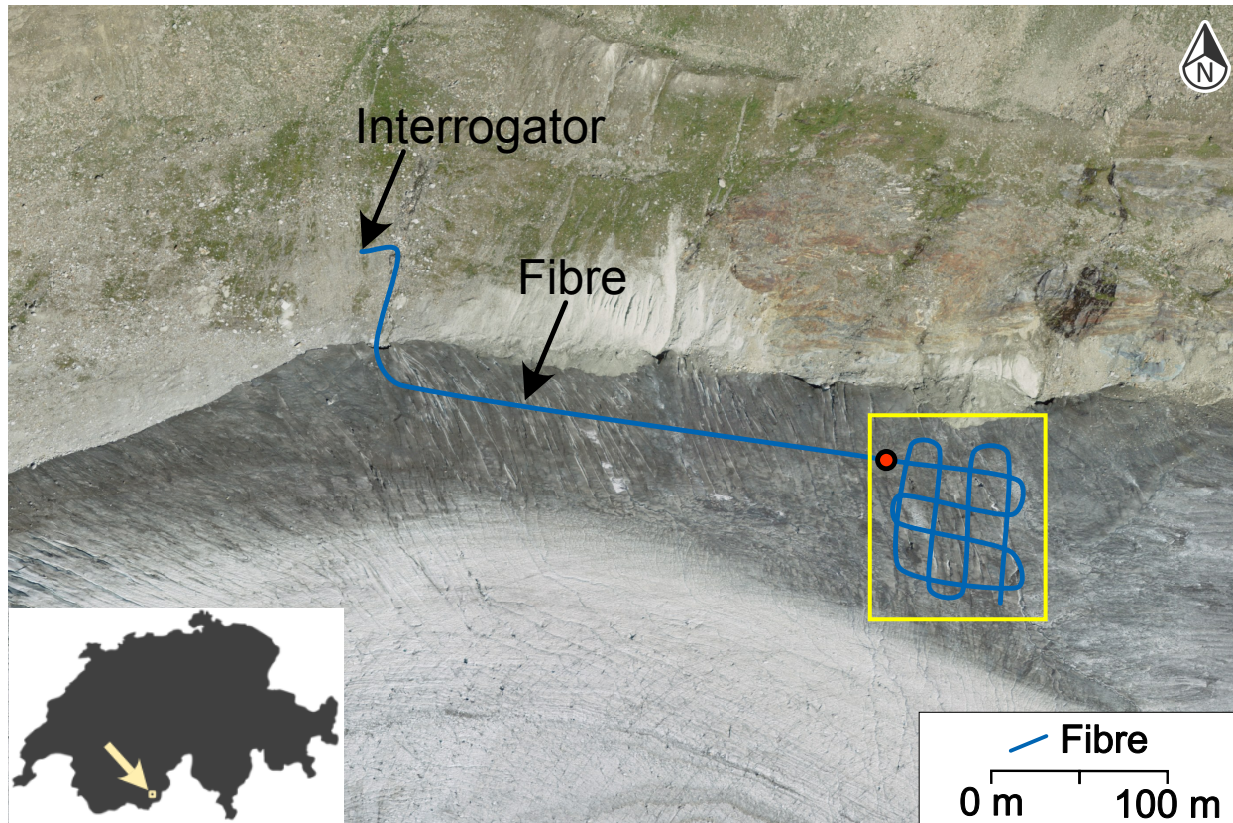


The dataset

Location: Gornergletscher, Swiss Alps

Fibre length: 1 km

Interrogator: Sintela Onyx (with 6 m gauge length, sampling @1000 Hz)

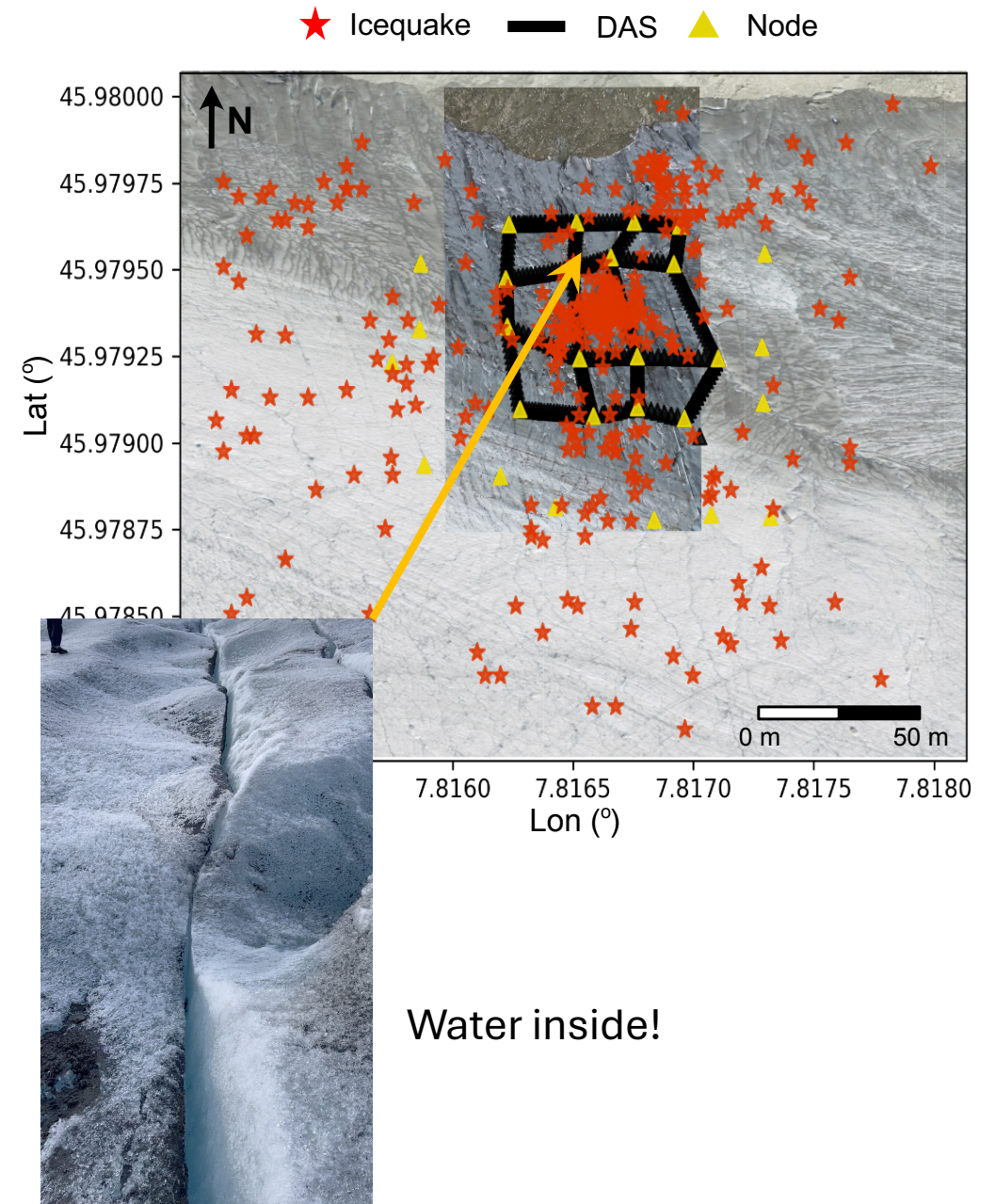
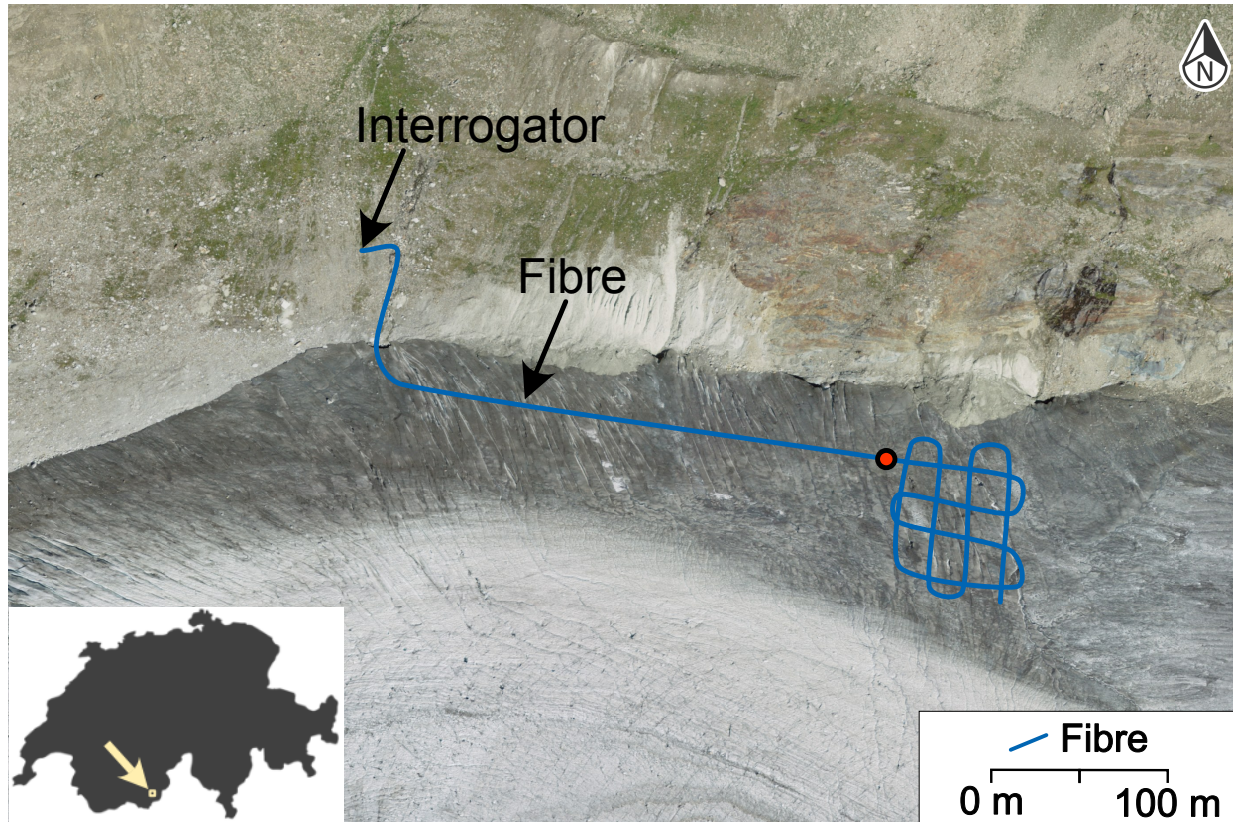


The experiment

Location: Gornergletscher, Swiss Alps

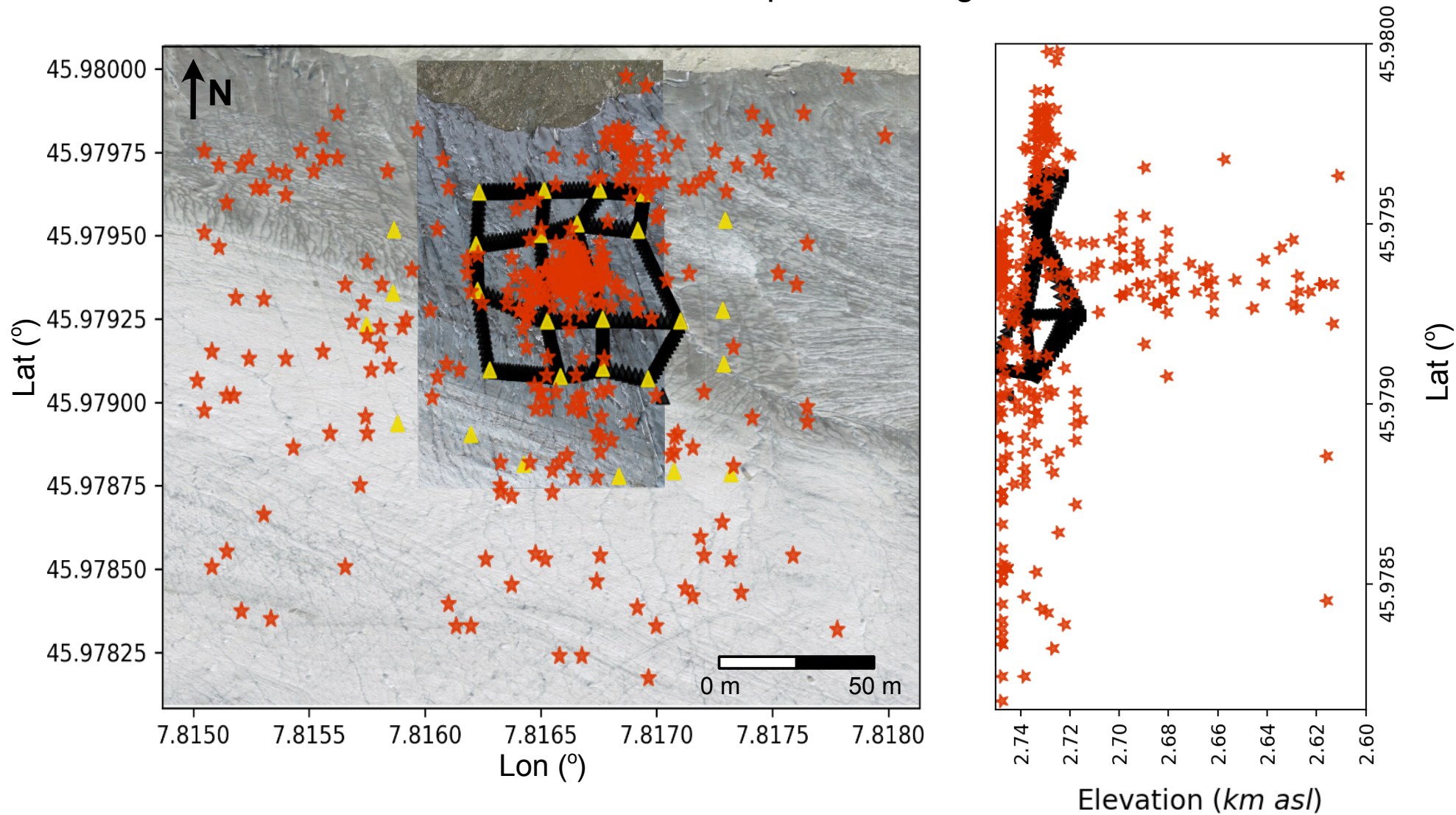
Fibre length: 1 km

Interrogator: Sintela Onyx (with 6 m gauge length, sampling @1000 Hz)

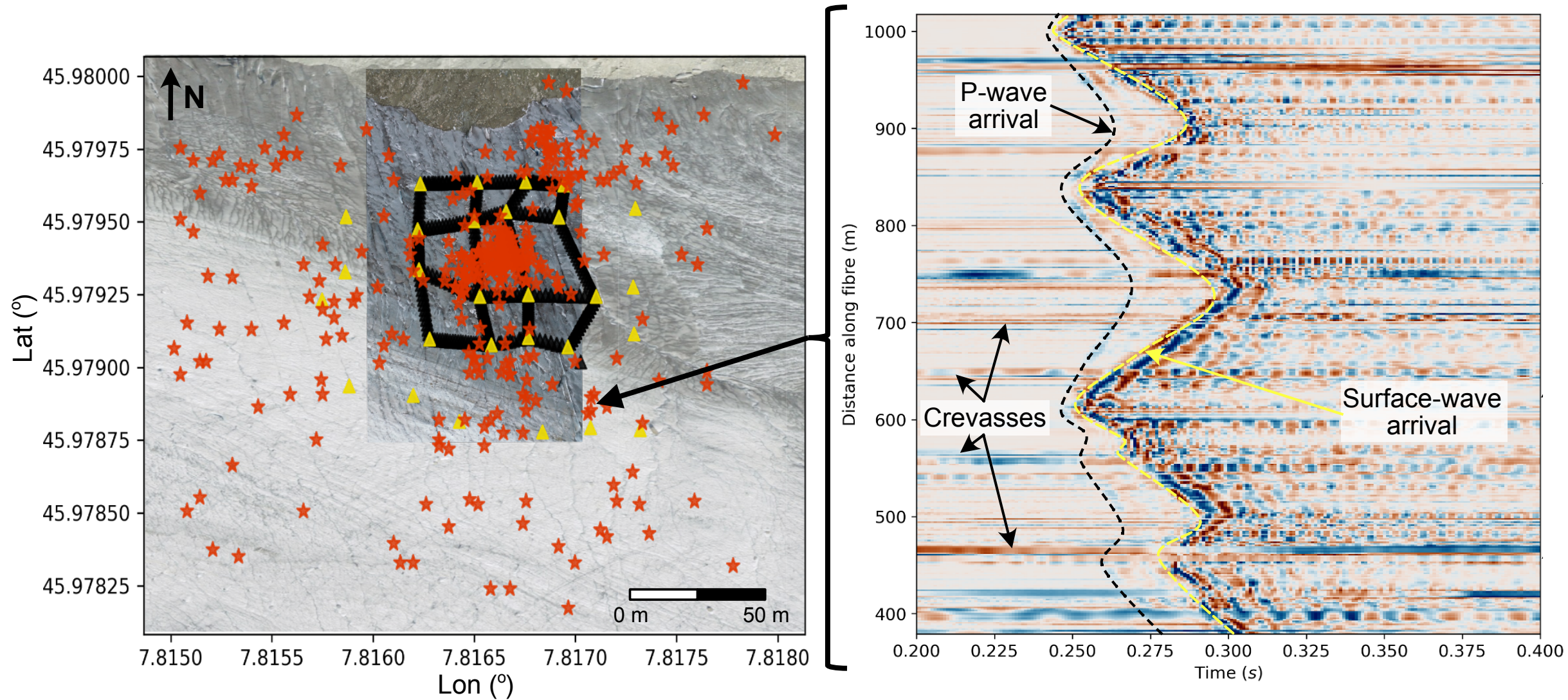


Detecting and locating seismicity

Icequake catalogue:



Example icequake



This practical:

Two options



Icequake source
mechanism inversion



Surface-wave velocity
anisotropy inversion

Seismic source mechanism inversion: Introduction

1. Source mechanisms (moment tensors)

2. Data

3. Inversion setup

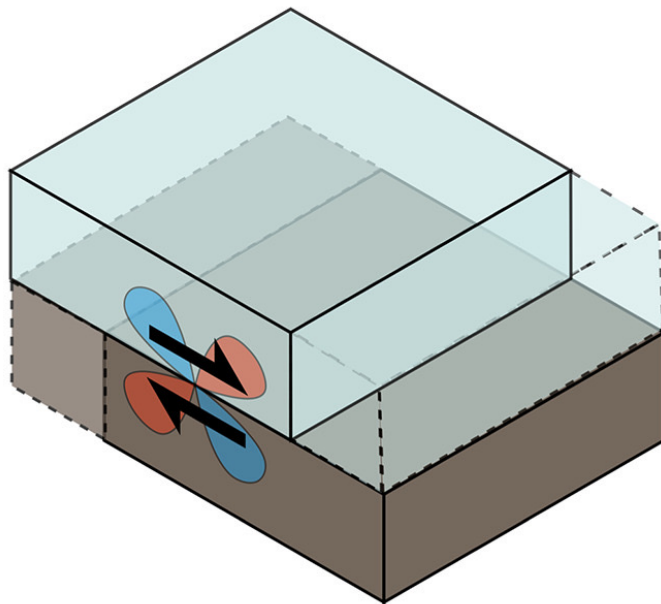
4. Nuances of fibreoptics

Alvizuri et al. (2018)

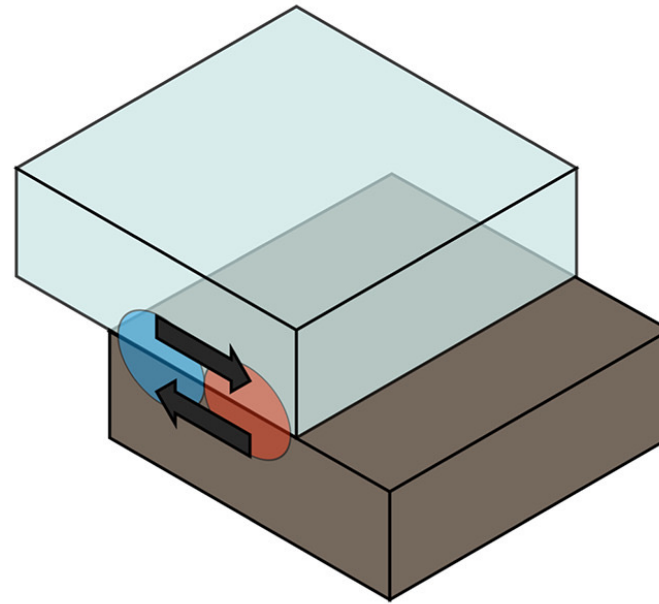
Source mechanisms

Fundamental types:

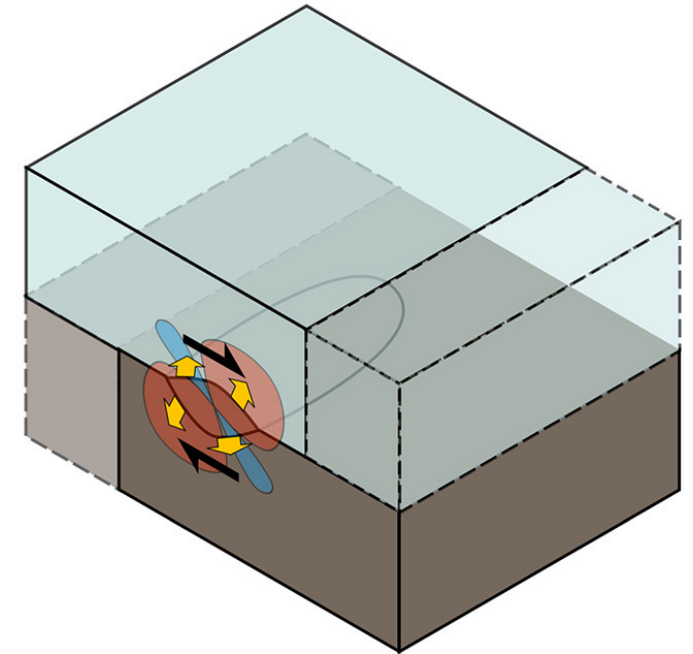
(a) Double-couple (DC)



(b) Single-force (SF)

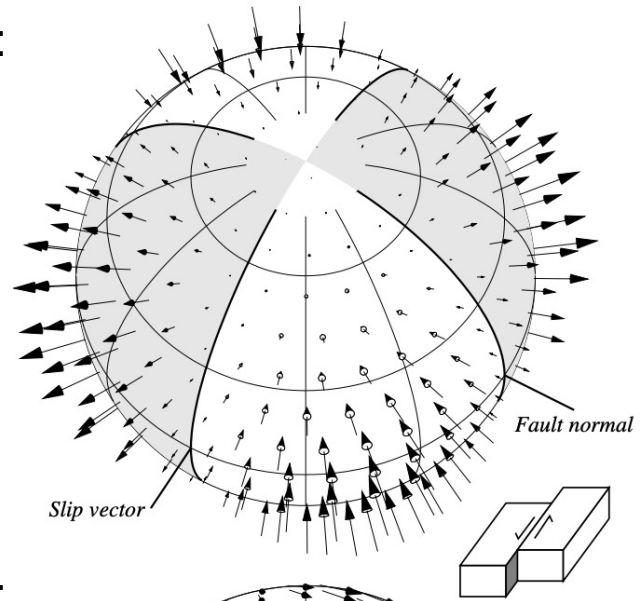


(c) Full moment tensor (MT)

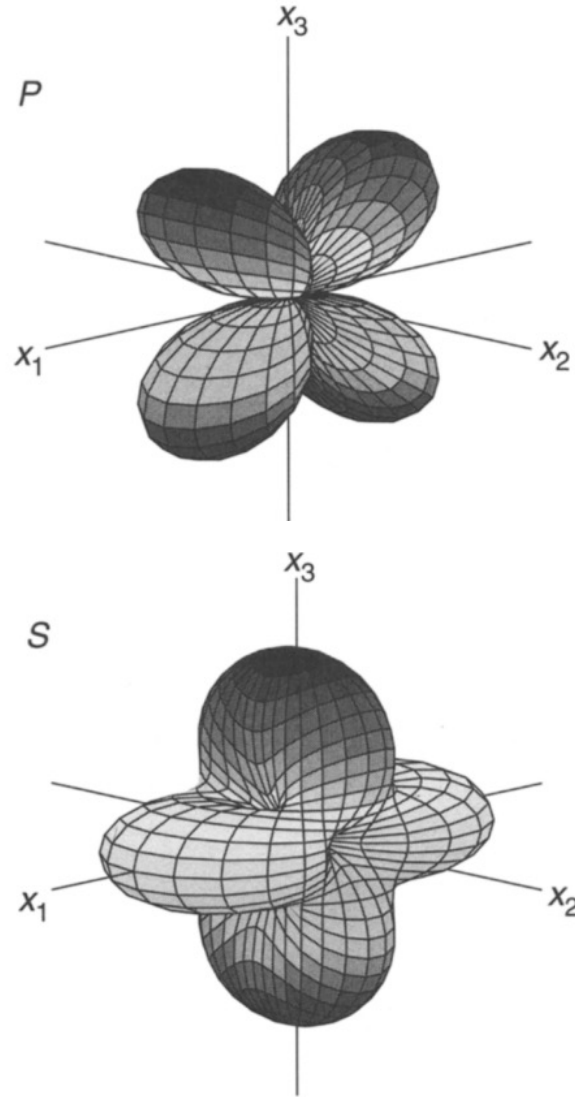
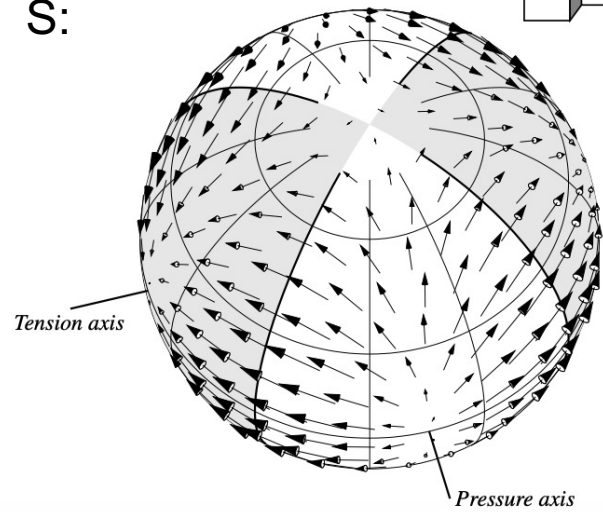


Radiation patterns

P:

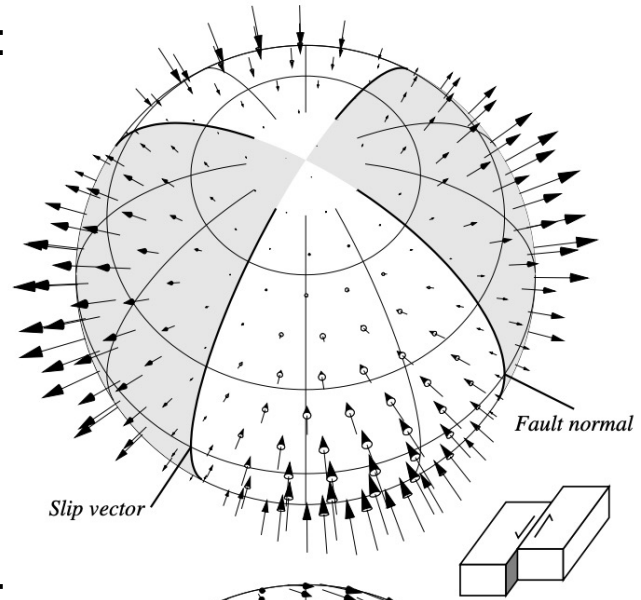


S:

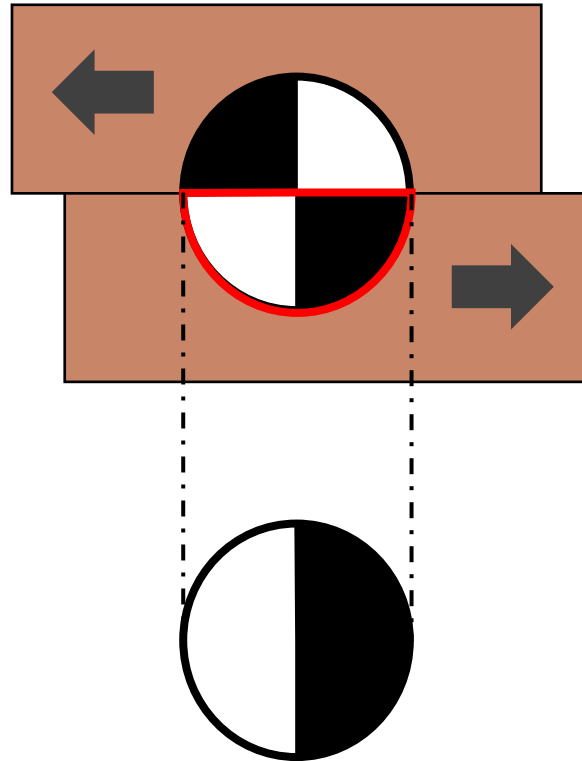
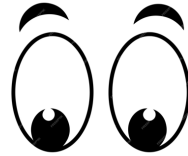
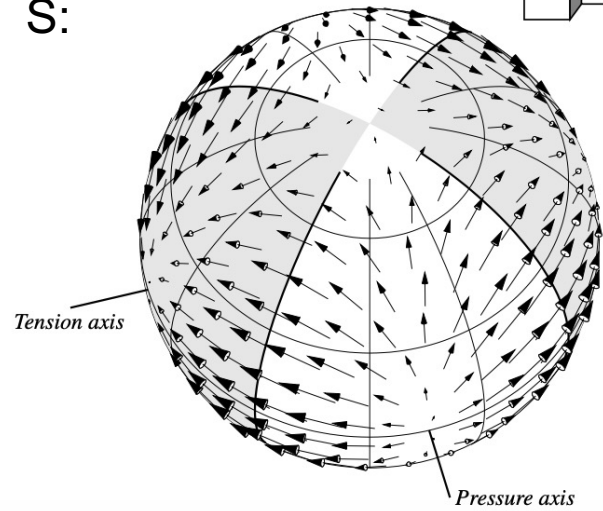


Radiation patterns

P:

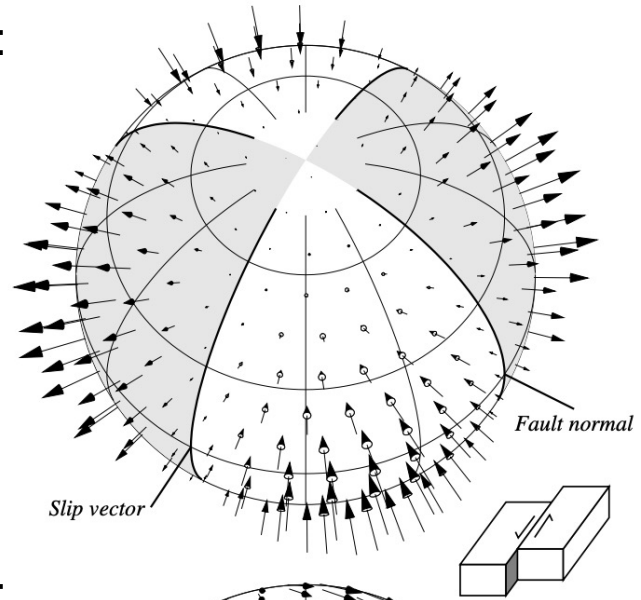


S:

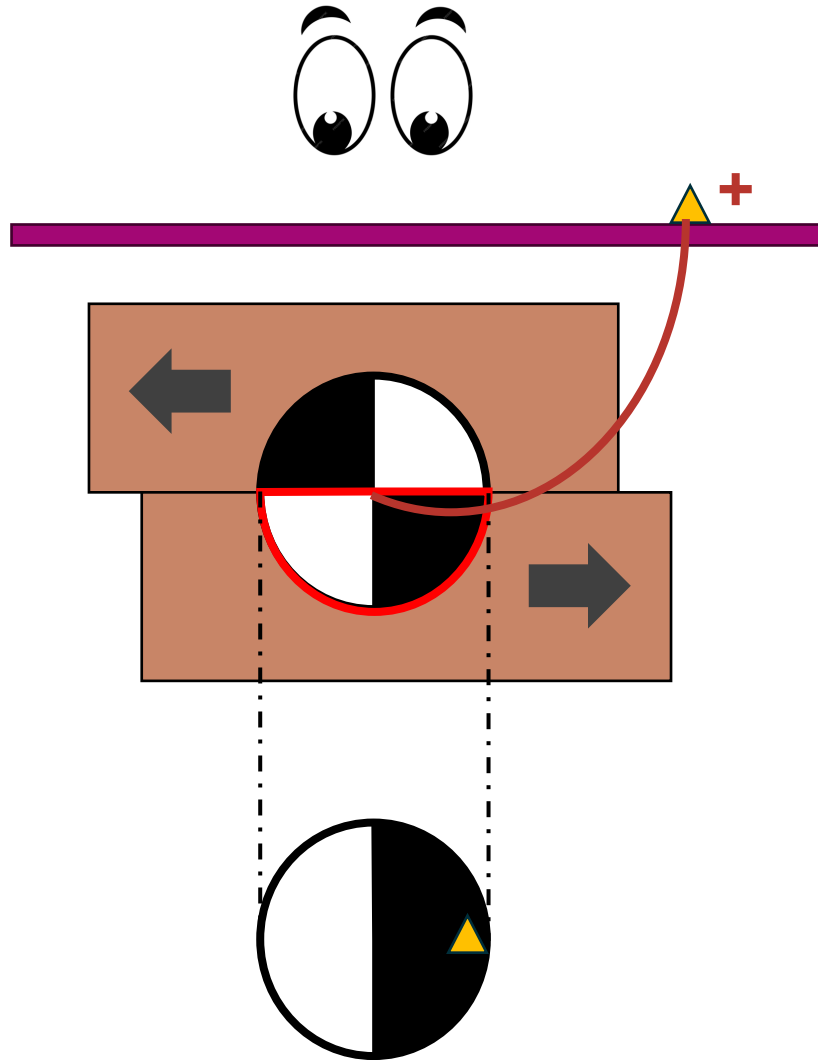
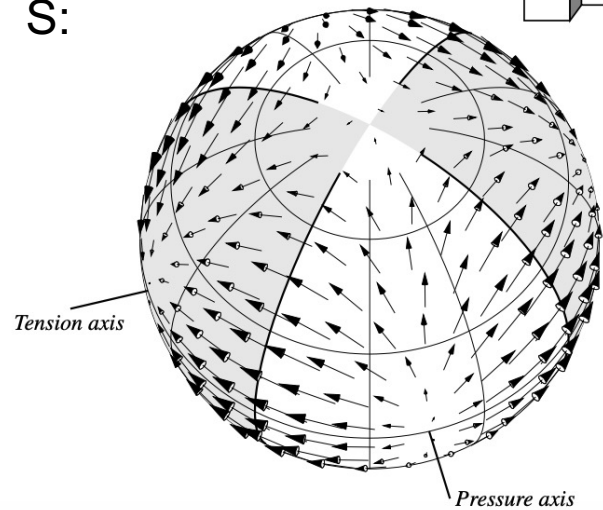


Radiation patterns

P:

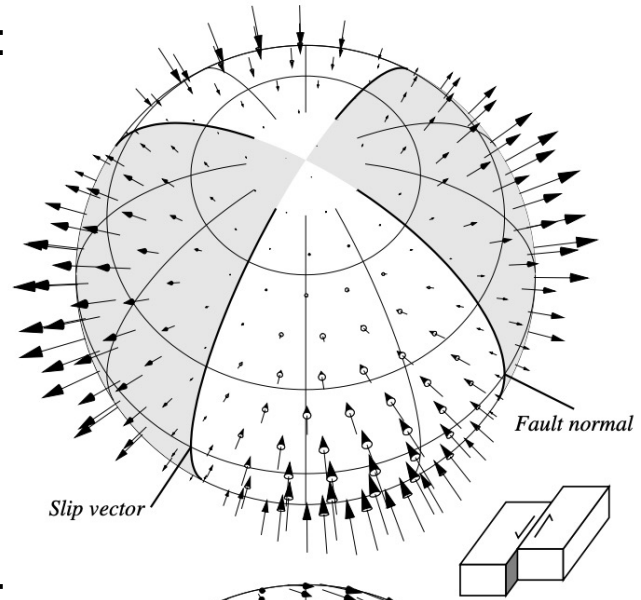


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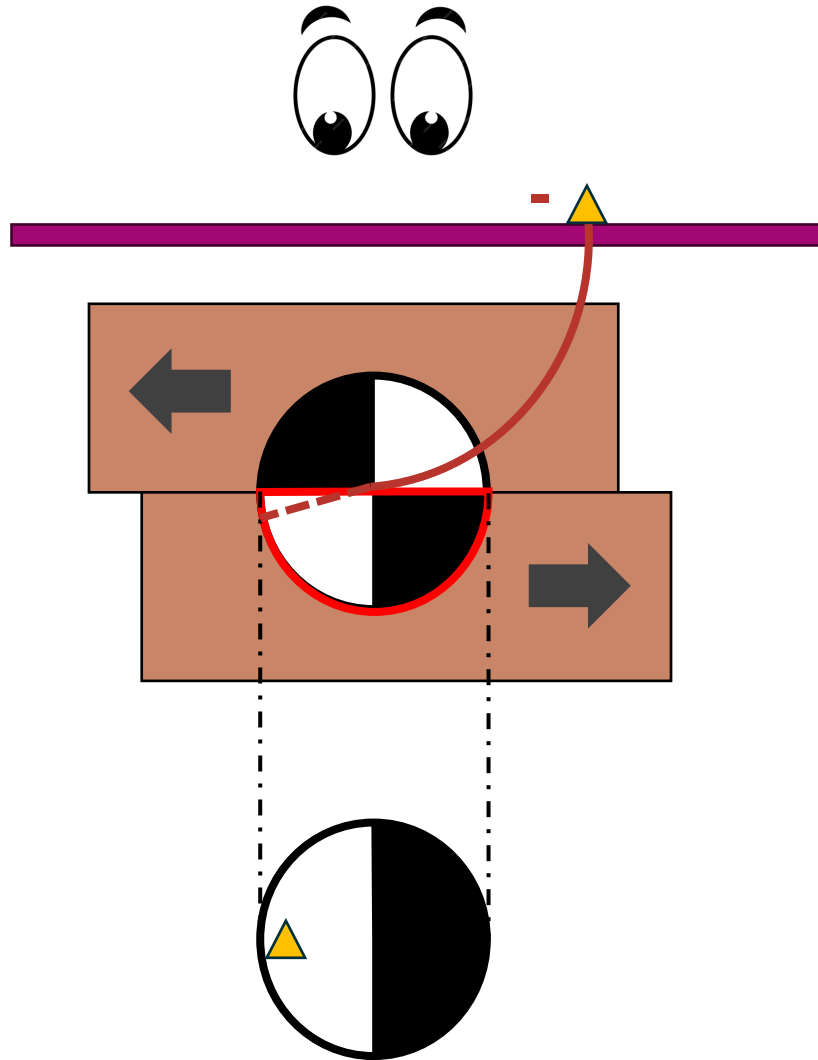
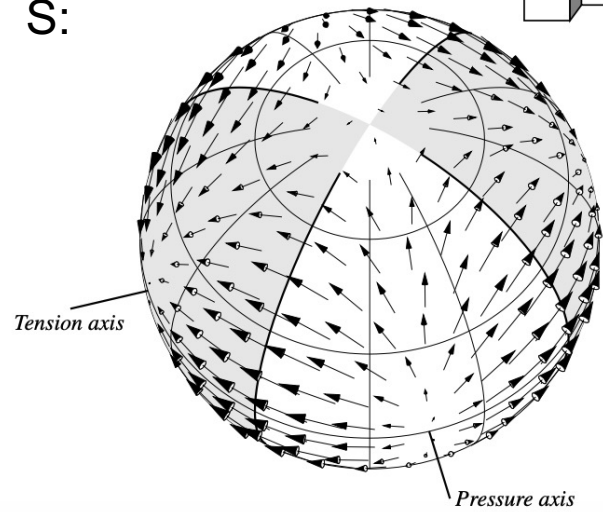


Radiation patterns

P:

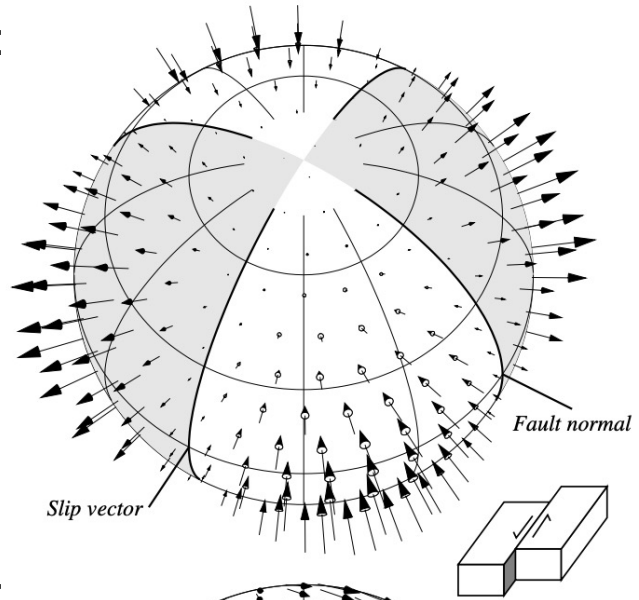


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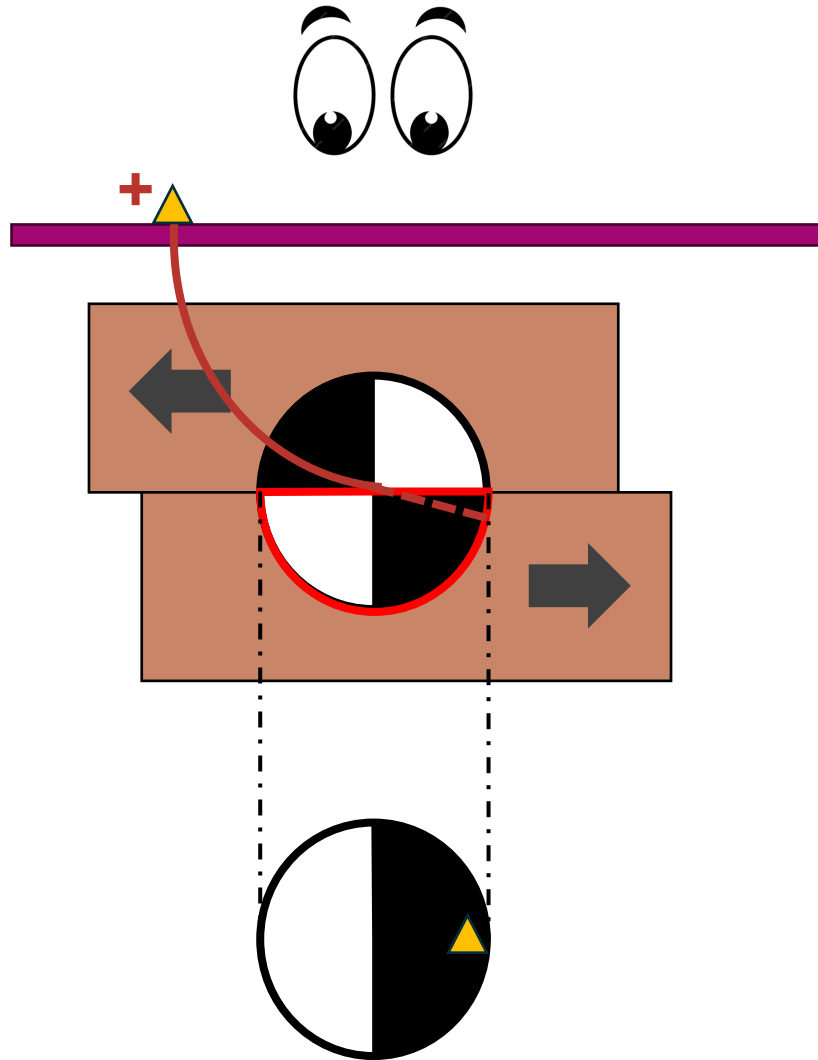
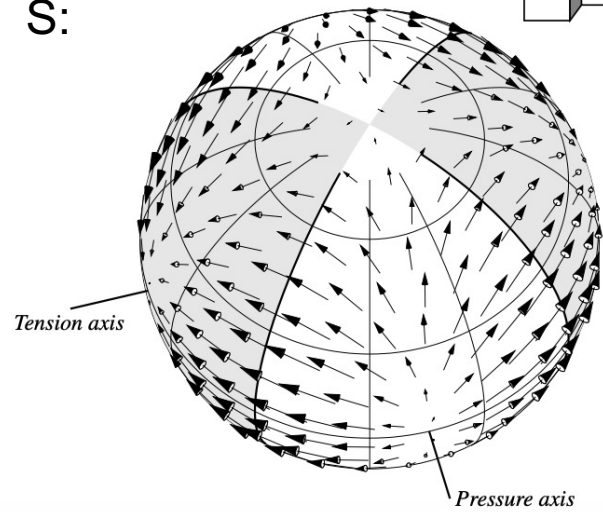


Radiation patterns

P:

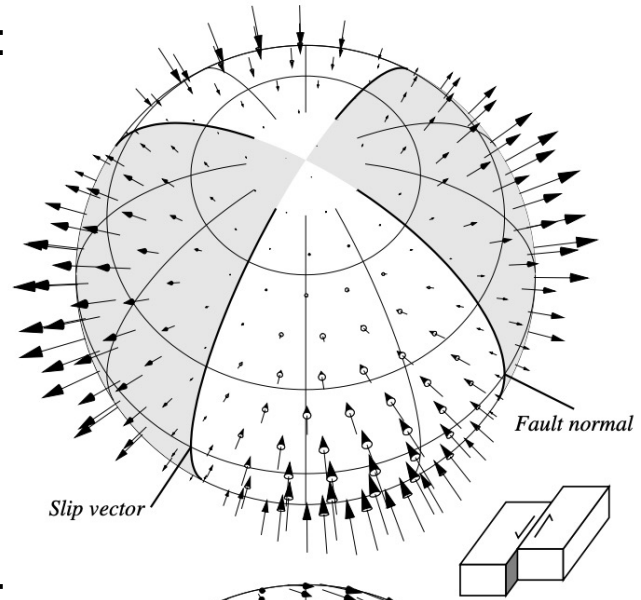


S:

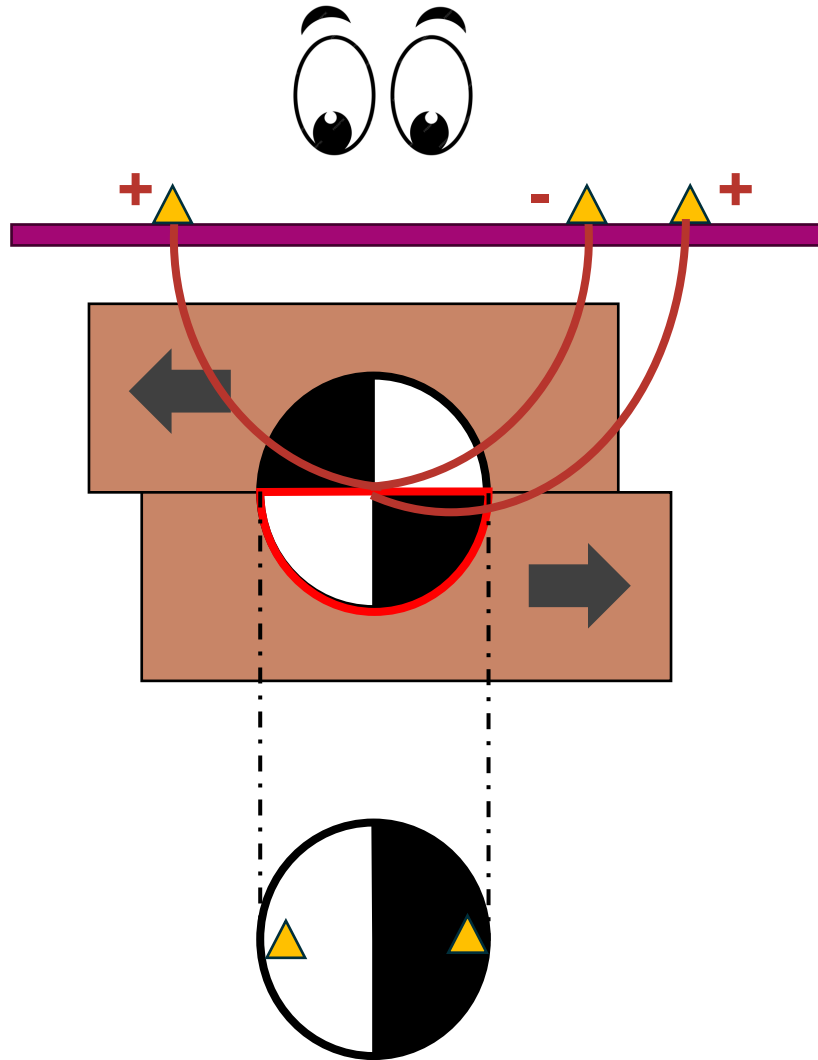
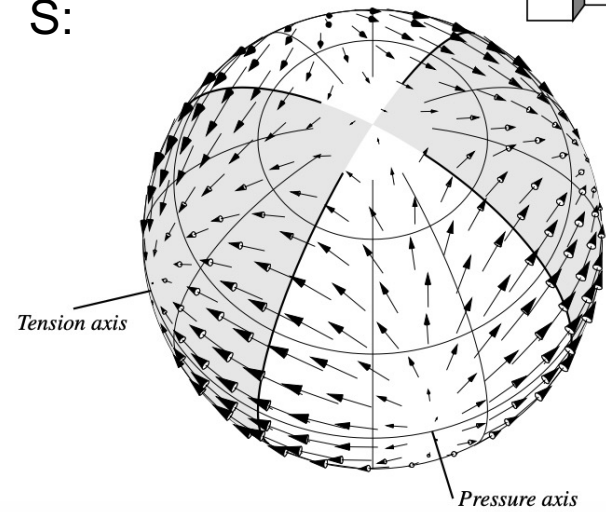


Radiation patterns

P:

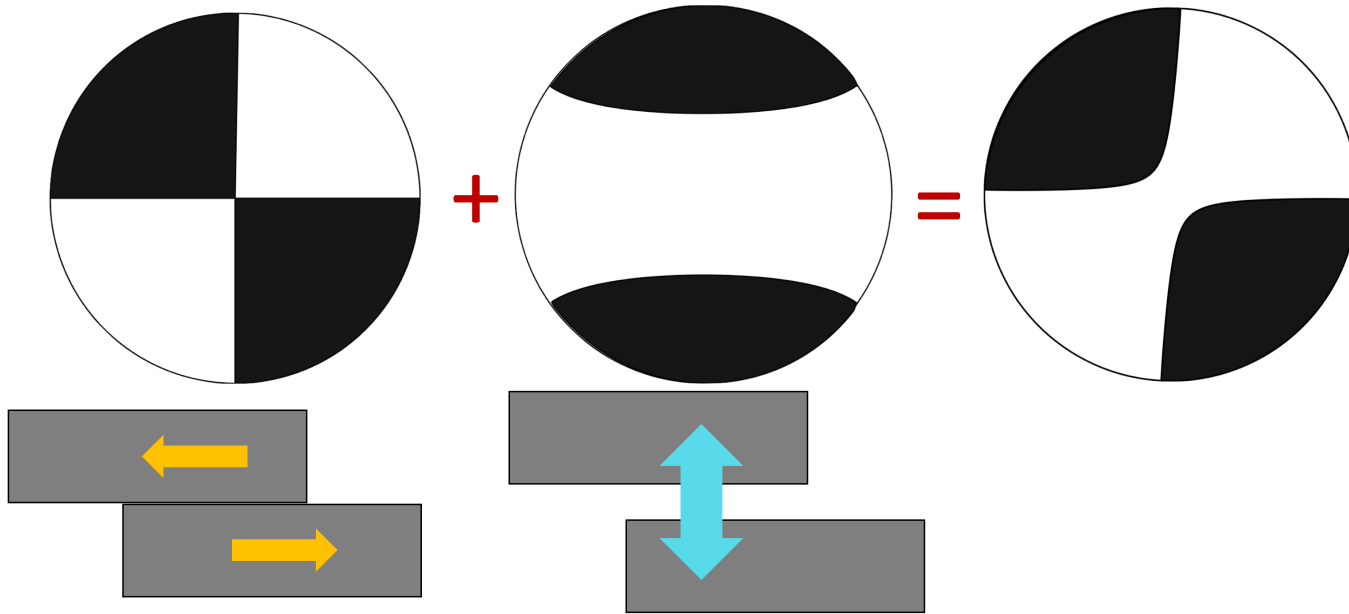


S:

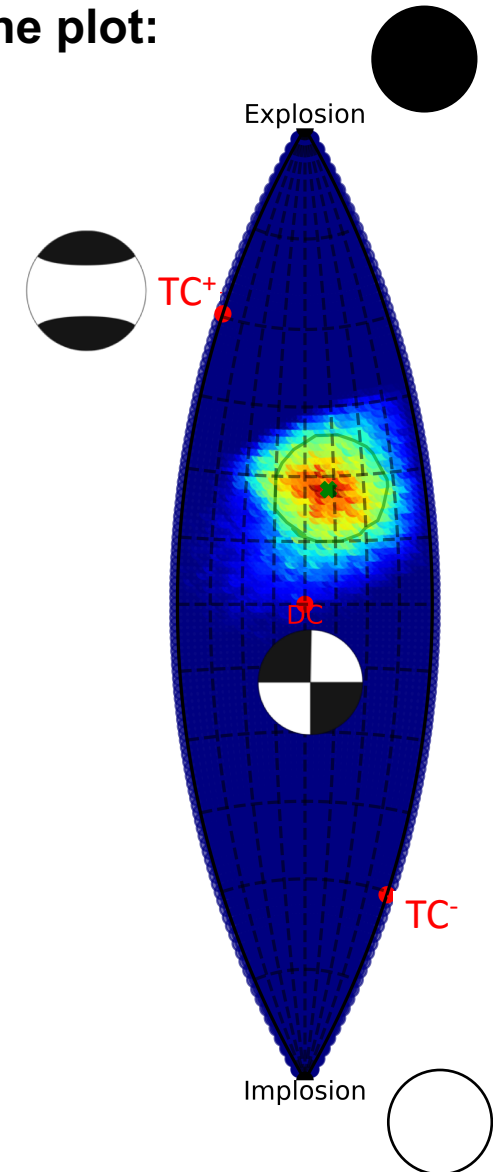


Radiation patterns

Non-double couple sources:



Lune plot:



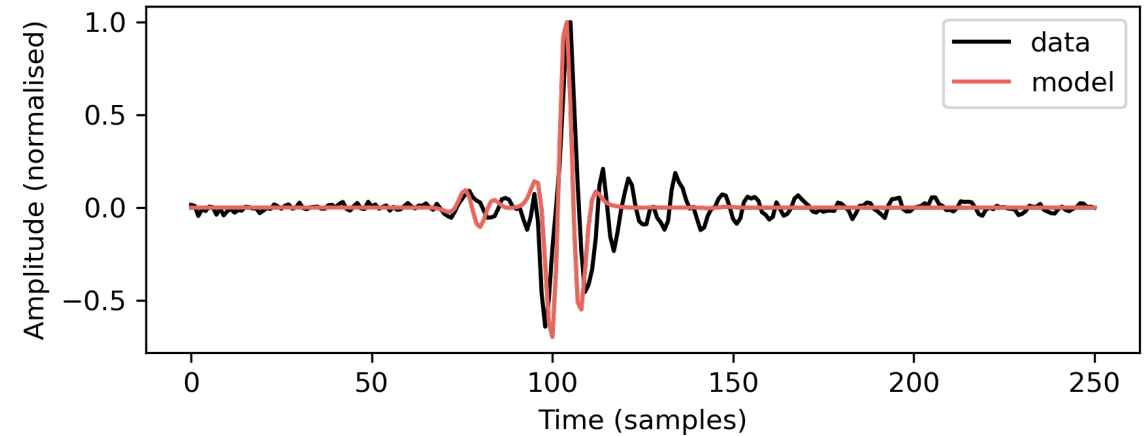
Seismic source mechanism inversion: Introduction

1. Source mechanisms (moment tensors)

2. Data

3. Inversion setup

4. Nuances of fibreoptics



Seismic source mechanism inversion: Introduction

1. Source mechanisms (moment tensors)

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Surface wave velocity anisotropy inversion: A little theory

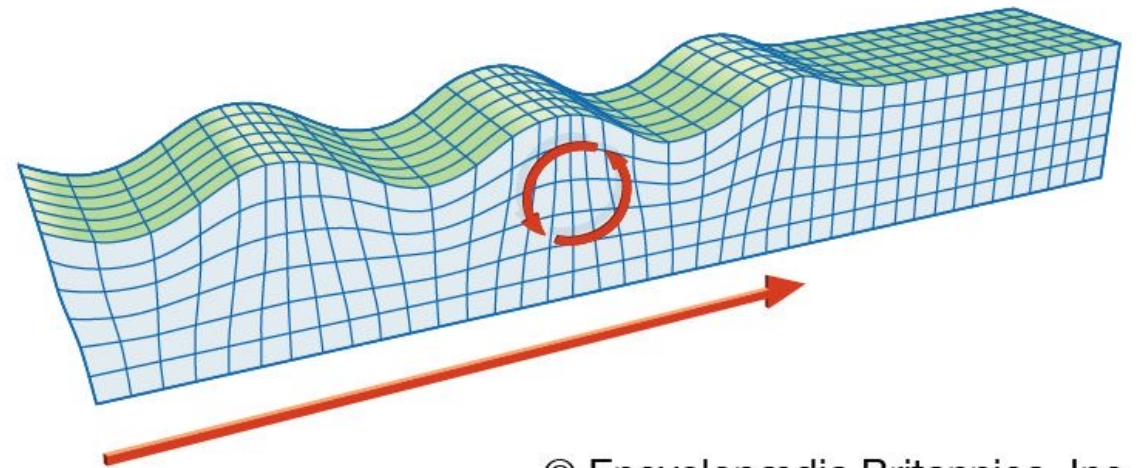
1. Rayleigh waves

2. Anisotropy

3. Data

4. Inversion setup

Rayleigh wave



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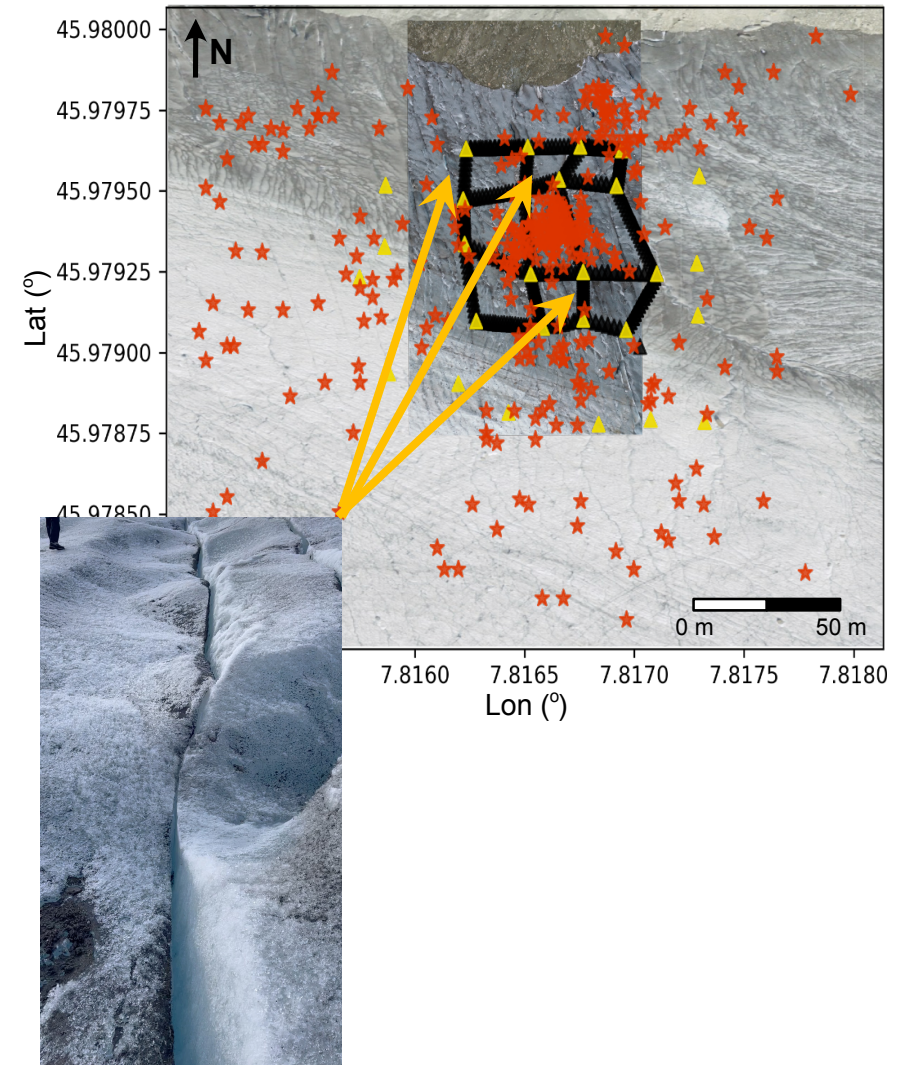
Surface wave velocity anisotropy inversion: A little theory

1. Rayleigh waves

2. Anisotropy

3. Data

4. Inversion setup



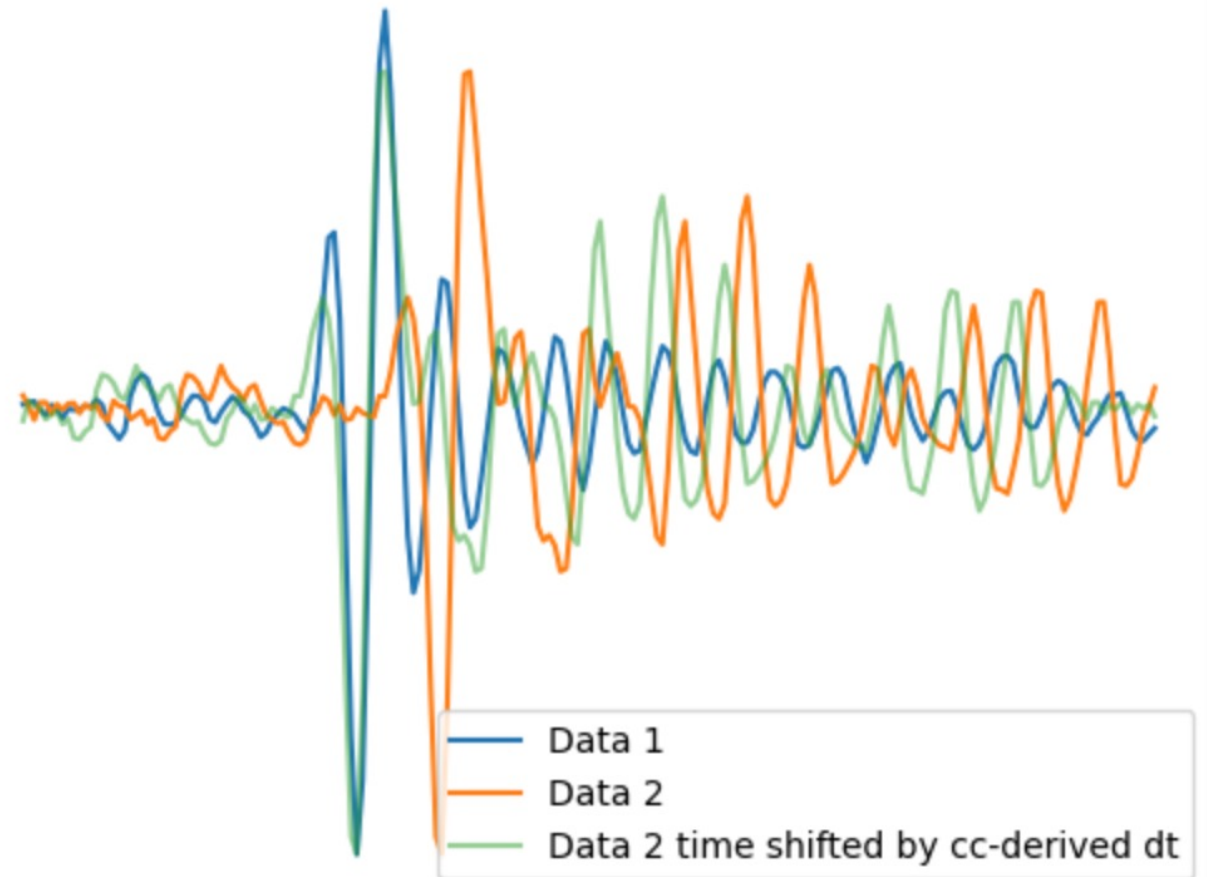
Surface wave velocity anisotropy inversion: A little theory

1. Rayleigh waves

2. Anisotropy

3. Data

4. Inversion setup



Surface wave velocity anisotropy inversion: A little theory

1. Rayleigh waves

2. Anisotropy

3. Data

4. Inversion setup

Seismic source mechanism inversion

Structure:

- Jupyter notebook:
 - Familiarisation with dataset
 - Setup inversion
 - Run inversion
 - * Play around with visualisation of results

Learning aims:

- Exposure to fibreoptic sensing data
- Introduction to linear inversion
- Introduction to full waveform seismic source inversion

Getting started:

- Copy data
- Install any dependencies (NonLinLocPy, SeisSrcInv)
- Open notebook

Surface wave velocity anisotropy inversion

Structure:

- Jupyter notebook:
 - Familiarisation with dataset
 - Setup inversion
 - Run inversion (synthetic + real data)
 - * Further develop yourselves

Learning aims:

- Exposure to fibreoptic sensing data
- Introduction to linear inversion (and a very simple approach to non-linear systems)
- Introduction to glacier anisotropy

Getting started:

- Copy data
- Install any dependencies (NonLinLocPy)
- Open notebook



Time to share ideas...



To conclude...

More familiar with fibreoptic sensing (DAS) data

Have had a fun play with a seismic inversion problem